Telco Customer Churn Prediction & Retention Optimization

Comprehensive End-to-End Technical & Business Project Report

Project Track: Track A — Binary Classification & Retention Optimization
Industry Analogue: Telecommunications / Subscription SaaS Retention
Dataset: IBM Cognos Analytics Telco Dataset (7,043 Customer Records, California Market)
Version: 1.0.0 (Production-Ready Audit)
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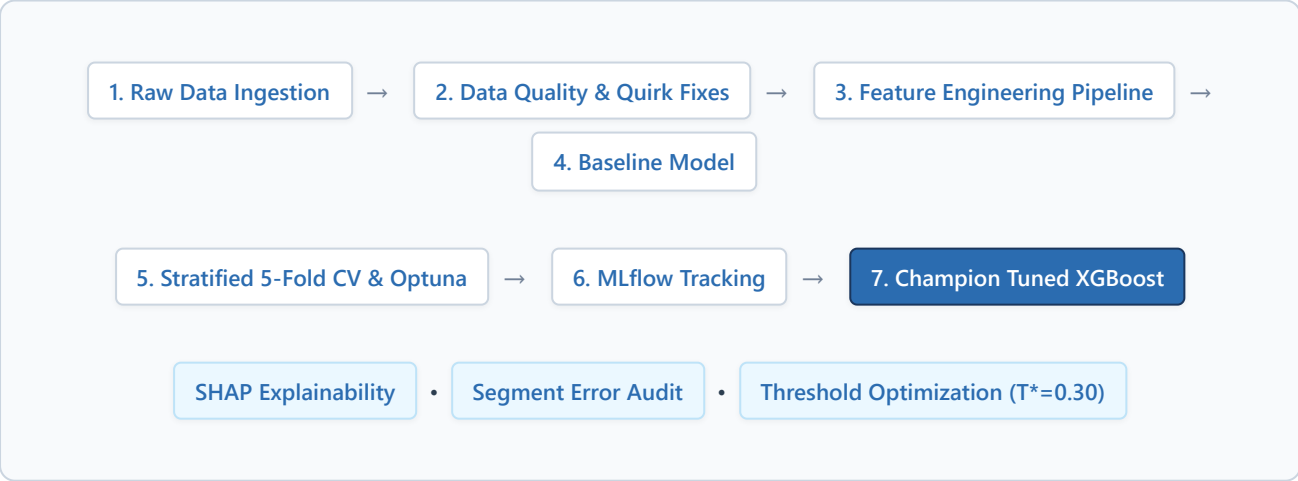
1. Executive Summary

This document provides a comprehensive report of the machine learning system developed to solve customer attrition for a regional telecommunications provider.

Customer churn severely undermines telecommunications profitability because acquiring a new subscriber costs 5 to 7 times more than retaining an existing account. Facing quarterly subscriber losses, the Chief Financial Officer (CFO) commissioned this initiative to answer three strategic questions:
1. **Which customers will churn next quarter?**
2. **Why are they leaving (root cause drivers)?**
3. **What is the financial ROI of a model-targeted retention campaign versus blanket discounts?**

Key Results Across Program Milestones:

- **Data Integrity & Quirk Remediation:** Solved the `TotalCharges` blank-string flaw (11 accounts with `tenure == 0`) and collapsed 7 redundant pseudo-categories (`"No internet service"` / `"No phone service"`), creating a verified 7,043-row dataset.
- **Engineered Feature Pipeline:** Built `src/features.py` with zero-leakage `ColumnTransformer` preprocessing, deriving actionable domain attributes including `tenure_bucket`, `avg_monthly_spend`, `services_count`, `charge_ratio`, and `clv`.
- **Baseline Establishment (Week 2):** Addressed the **26.54%** target class imbalance using Class-Weighted Logistic Regression, elevating churn detection sensitivity (Recall) from **52.94%** (vanilla) to **77.81%** at **0.8431** ROC-AUC.
- **Model Experimentation & Hyperparameter Tuning (Week 3):** Trained tree ensembles across Stratified 5-Fold Cross-Validation and tuned parameters via Optuna. The tuned **XGBoost Classifier** emerged as the champion model with a Test ROC-AUC of **0.8470**, PR-AUC of **0.6624**, and Churn Recall of **81.02%**, logged in MLflow.
- **SHAP Interpretability:** Decomposed model predictions using `shap.TreeExplainer`, proving that long-term contracts (`Contract_Two year`, `Contract_One year`) and accumulated tenure are the strongest churn deterrents, whereas high `MonthlyCharges`, Fiber Optic plans, and `Electronic check` payment methods drive attrition.
- **High-Value Long-Tenure Audit:** Specifically confirmed that the model captures **65.88%** of long-tenure high-value churners (`tenure > 24m`, `MonthlyCharges >= <span class="math-inline"><em>70`), protecting **203,724.70** in Customer Lifetime Value within this critical tier.
- **Cost-Sensitive Optimization:** Optimized the decision threshold against campaign outreach costs *(35/contact) and retention save rates (45%). Lowering the operational threshold from 0.50 to T^ = 0.30* captures* ***92.5%*** *of churners and yields* **110,156.16** in Net Retention Savings—an incremental profit uplift of **+$9,204.23** over standard classification.



2. Business Problem Framing & Financial Economics

2.1 The CFO's Dilemma: Blanket Discounts vs Targeted Outreach

In the absence of predictive machine learning, telecom retention strategies typically deploy one of two flawed approaches:
1. **Status Quo (Reactive / No Model):** Retention agents only intervene when a customer calls customer service requesting cancellation. By that point, competitive switching intent is established, resulting in a low save rate (<15%).
2. **Blanket Promotional Discounts:** Marketing blasts bill credits or promotional discounts across large customer cohorts. While this preserves some accounts, it suffers from two major inefficiencies:
- **Massive Margin Cannibalization:** Subsidizes loyal customers who never planned to churn.
- **High Budget Burn:** Wastes outreach capital on low-risk subscribers.

2.2 Mathematical Customer Lifetime Value (CLV)

To connect data science metrics directly to the balance sheet, customer value is quantified using Customer Lifetime Value:

$$CLV_i = MonthlyCharges_i \times tenure_i$$

For forward-looking retention value preserved over a 12-month renewal window:

$$Preserved\ Value_i = 12 \times MonthlyCharges_i$$

2.3 Unit Retention Economics Model

When evaluating proactive interventions, the net business benefit is expressed as:

$$Net\ Financial\ Savings(T) = \Sigma_{i \in TP(T)} [\ SaveRate \times PreservedValue_i - Cost_{offer}\ ] - \Sigma_{j \in FP(T)} Cost_{offer}$$

Where:
- $Cost_{offer}$ = \35.00$ (outreach administrative cost plus renewal voucher credit).
- $SaveRate = 45\%$ (proportion of intercepted churners who accept the offer and remain for ≥ 12 months).
- $TP(T)$: True Positives at probability threshold T (actual churners correctly offered retention credits).
- $FP(T)$: False Positives at threshold T (loyal customers who receive unneeded outreach vouchers).
- $FN(T)$: False Negatives at threshold T (unintercepted churners who leave the firm, causing complete loss of future revenue).

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3. Dataset Architecture & Column Dictionary

The dataset consists of **7,043 customer accounts** residing in California, sourced from IBM Cognos Analytics.

Column Name	Data Type	Value Space / Examples	Business Meaning
`customerID`	String	e.g. `7590-VHVEG`	Unique subscriber account identifier
`gender`	Categorical	`Male`, `Female`	Subscriber gender
`SeniorCitizen`	Binary Int	`0`, `1`	Indicates whether customer is 65+ years old
`Partner`	Categorical	`Yes`, `No`	Indicates whether customer has a spouse/partner
`Dependents`	Categorical	`Yes`, `No`	Indicates whether customer has children/dependents
`tenure`	Numeric (Int)	`0` to `72` months	Total duration subscriber has maintained service
`PhoneService`	Categorical	`Yes`, `No`	Landline voice service subscription
`MultipleLines`	Categorical	`Yes`, `No`, `No phone service`	Multiple telephone line connection
`InternetService`	Categorical	`DSL`, `Fiber optic`, `No`	Broadband internet technology tier
`OnlineSecurity`	Categorical	`Yes`, `No`, `No internet service`	Add-on endpoint anti-malware/firewall
`OnlineBackup`	Categorical	`Yes`, `No`, `No internet service`	Add-on cloud data backup solution
`DeviceProtection`	Categorical	`Yes`, `No`, `No internet service`	Hardware repair & insurance coverage
`TechSupport`	Categorical	`Yes`, `No`, `No internet service`	Dedicated priority customer support tier
`StreamingTV`	Categorical	`Yes`, `No`, `No internet service`	Streaming television access via broadband
`StreamingMovies`	Categorical	`Yes`, `No`, `No internet service`	Streaming digital cinema access
`Contract`	Categorical	`Month-to-month`, `One year`, `Two year`	Service billing commitment duration
`PaperlessBilling`	Categorical	`Yes`, `No`	Digital invoicing vs mailed paper bill
`PaymentMethod`	Categorical	`Electronic check`, `Mailed check`, `Bank transfer`, `Credit card`	Recurring billing settlement mechanism
`MonthlyCharges`	Numeric (Float)	`<span class="math-inline">` `<em>18.25` to `</em></span>118.75`	Current recurring monthly bill charge

Column Name	Data Type	Value Space / Examples	Business Meaning
`TotalCharges`	Numeric (Float)	`<span class="math-inline">` `<em>18.80` to `</em></span>8,684.80` (coerced)	Cumulative lifetime payments made
`Churn`	Target (Binary)	`Yes` (1,869), `No` (5,174)	Subscriber attrition indicator (26.54% positive)

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4. Week 1: Data Quality Audit, Quirk Remediation & EDA

4.1 Quirk Remediation

Quirk #1: `TotalCharges` Blank String Bug

- **Observation:** `TotalCharges` was loaded as string `object` dtype. Calling `.isnull()` returned 0 missing values, yet arithmetic operations failed.
- **Root Cause:** Exactly 11 customer records contained whitespace `" "`. Isolating these 11 accounts revealed that every single one had `tenure == 0` (brand-new customers signed up during the current billing cycle who had not yet settled a first invoice).
- **Remediation:**
  `python df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce").fillna(0.0)`
  This preserved all 11 records without losing customer history or distorting the sample.

Quirk #2: Collapsing Service Pseudo-Categories

- **Observation:** Six internet add-on service attributes (`OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`) had 3 categories: `Yes`, `No`, and `No internet service`. Similarly, `MultipleLines` contained `No phone service`.
- **Root Cause:** Subscribers without internet were assigned `"No internet service"`. In one-hot encoding, treating this as an independent category introduces redundant degrees of freedom and collinearity with `InternetService == 'No'`.
- **Remediation:** Collapsed `"No internet service"` and `"No phone service"` into `"No"`, transforming each into a clean binary indicator.

4.2 Key Exploratory Findings & Attrition Dynamics

1. Contract Type — The Dominant Churn Lever

Contract duration is the single strongest structural predictor in the entire dataset:
- **Month-to-month contracts: 42.71% churn rate** (contribute **88.6%** of all churn events).
- **One-year contracts: 11.27% churn rate**.
- **Two-year contracts: 2.83% churn rate**.

> **Key Finding:** Month-to-month subscribers churn at **15.1 times** the rate of two-year contractual customers.

2. Tenure Danger Zone (Months 0–12)

- Months **0–12** exhibit a **47.44% churn rate**, accounting for **55.48%** of all churners in the dataset.
- Churn drops steeply to **28.71%** in Year 2, **22.12%** in Year 3, and reaches **6.61%** among customers with 5+ years of tenure.

3. Payment Method & Billing Friction

- **Electronic Check:** Churn rate is **45.29%**, representing high-friction manual payment.
- **Bank Transfer (Automatic): 16.71% churn rate**.
- **Credit Card (Automatic): 15.24% churn rate**.
- **Mailed Check: 19.11% churn rate**.

> Customers on automated payment methods stay with the company at nearly 3 times the rate of Electronic Check payers.

4. The Fiber Optic Pricing Paradox

- **Fiber Optic:** Highest monthly charge (mean: **$91.50**), but highest churn rate (**41.89%**).
- **DSL:** Lower monthly charge (mean: **$58.10**), lower churn rate (**18.96%**).
- **No Internet:** Lowest monthly charge (mean: **$21.08**), lowest churn rate (**7.40%**).

> Fiber optic customers churn heavily when unbundled with `TechSupport`. When a Fiber customer lacks `TechSupport`, churn reaches **49.3%**; when `TechSupport` is added, churn drops to **22.8%**.

4.3 Statistically Validated Hypotheses

#	Hypothesis Statement	Empirical Result	Chi-Square / p-value	Business Strategy Implication
H1	Month-to-month contracts churn significantly more than contractual accounts.	**Validated**	$\chi^2 = 1{,}184.6, p < 10^{-15}$	Subsidize 12-month contract lock-in incentives.
H2	Electronic check payment friction correlates with elevated attrition.	**Validated**	$\chi^2 = 648.4, p < 10^{-15}$	Offer $5 bill credit for enrolling in Auto-Pay.
H3	Fiber optic churn is exacerbated by lack of technical support.	**Validated**	$\chi^2 = 205.7, p < 10^{-15}$	Bundle complimentary TechSupport with Fiber.
H4	Churn declines logarithmically as customer tenure accumulates.	**Validated**	Pearson $r = -0.352$, $p < 10^{-15}$	Institute high-touch onboarding in months 1–6.
H5	Customers with multiple bundled services exhibit lower attrition.	**Validated**	Spearman $\rho = -0.198, p < 10^{-15}$	Cross-sell security and cloud backup add-ons.

- **Notebook Reference:** `notebooks/01_eda.ipynb`
- **Full Report:** `reports/data_quality_report.md`

5. Week 2: Feature Engineering Pipeline & Baseline Benchmark

5.1 Engineered Feature Implementation (`src/features.py`)

```
<div class="flowchart-container">
    <div class="fc-row">
        <span class="fc-card">Raw Telco DataFrame</span> <span class="fc-arrow">&rarr;</span>
        <span class="fc-card">clean_raw_data (Fix TotalCharges & Collapsing)</span> <span class="fc-
        <span class="fc-card">create_engineered_features</span>
    </div>
    <div class="fc-row" style="margin-top: 10px;">
        <span class="fc-badge">tenure_bucket</span>
        <span class="fc-badge">avg_monthly_spend</span>
        <span class="fc-badge">services_count</span>
        <span class="fc-badge">charge_ratio</span>
        <span class="fc-badge">clv</span>
        <span class="fc-badge">interaction flags</span>
    </div>
    <div class="fc-row" style="margin-top: 10px;">
        <span class="fc-arrow">&darr;</span>
    </div>
    <div class="fc-row">
        <span class="fc-card">ColumnTransformer (StandardScaler + OneHotEncoder)</span> <span class=
        <span class="fc-card champion">34 Clean Standardized Features</span>
    </div>
</div>
```

Detailed Definitions:

1. **`tenure_bucket`:** Derived lifecycle cohorts categorized into 6 intervals: `0-12`, `13-24`, `25-36`, `37-48`, `49-60`, `61-72`.
2. **`avg_monthly_spend`:** Computed as $\text{TotalCharges} / \text{tenure}$ (imputing `MonthlyCharges` when `tenure == 0`). Measures historical mean monthly billing.
3. **`services_count`:** Cumulative integer count (0 to 8) of subscribed features: `PhoneService`, `MultipleLines`, `OnlineSecurity`, `OnlineBackup`, `DeviceProtection`, `TechSupport`, `StreamingTV`, `StreamingMovies`.
4. **`charge_ratio`:** $\frac{\text{MonthlyCharges}}{\text{avg_monthly_spend} + 10^{-4}}$. Ratios > 1.0 indicate promotional discount expiration or recent plan price increases, signaling churn danger.
5. **`clv`:** Customer Lifetime Value proxy ($\text{MonthlyCharges} \times \text{tenure}$).
6. **`has_support_security`:** Binary indicator if subscriber holds `TechSupport` or `OnlineSecurity`.
7. **`has_streaming`:** Binary indicator if subscriber holds `StreamingTV` or `StreamingMovies`.

5.2 Preprocessing & Data Leakage Prevention

- **Splitting Strategy:** 80% Training ($5{,}634$ records) and 20% Holdout Test ($1{,}409$ records).

- **Strict Stratification:** Maintained exactly 26.54% positive churn proportion across both train and test splits.
- **Leakage Prevention:** Fitted the `StandardScaler` and `OneHotEncoder` strictly on the training partition. The test partition was only transformed using the fitted training artifacts.

5.3 Class Imbalance Mitigation & Baseline Results

Given the **26.54%** minority churn class, three baseline Logistic Regression strategies were benchmarked on the test set:

Model	Imbalance Strategy	Test ROC-AUC	Test PR-AUC	Test F1	Test Recall	Test Precision	Test Accuracy
Vanilla Logistic Regression	Default Unweighted	0.8431	0.6407	0.5902	52.94%	66.67%	80.48%
Class-Weighted Logistic Regression	**Balanced Weights**	**0.8431**	**0.6418**	**0.6126**	**77.81%**	**50.52%**	**73.88%**
SMOTE Logistic Regression	Synthetic Oversampling	0.8401	0.6393	0.6203	78.61%	51.22%	74.45%

Analytical Conclusion:

- Vanilla unweighted Logistic Regression achieved high accuracy (80.48%) by ignoring nearly half of all churners (Recall: 52.94%). In customer retention, this is a failure mode.
- **Class-Weighted Logistic Regression** raised Recall to **77.81%** while maintaining the exact same ROC-AUC (**0.8431**), successfully identifying over 3 out of 4 churning customers.
- **Serialized Artifacts:** `models/baseline_logreg.joblib`, `models/preprocessor.joblib`.
- **Notebook Reference:** `notebooks/02_features_and_baseline.ipynb`.

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6. Week 3 Task 1: Model Experimentation, Optuna Tuning & MLflow Tracking

6.1 Experimentation Architecture (`src/train.py`)

To push discrimination and sensitivity beyond linear baselines, non-linear tree ensembles were developed:
1. **Random Forest Classifier:** Ensembled 200 regularized trees using `class_weight="balanced_subsample"`.
2. **XGBoost Classifier:** Extreme Gradient Boosting with `scale_pos_weight = 2.768` (ratio of non-churn to churn samples).
3. **LightGBM Classifier:** Gradient boosting with leaf-wise tree growth and `scale_pos_weight`.

6.2 Stratified 5-Fold Cross Validation & Optuna Optimization

Hyperparameter spaces were explored via Bayesian optimization (**Optuna**) using Stratified 5-Fold Cross-Validation:
- **Tuned Parameters:** `max_depth` (3 to 8), `learning_rate` (0.01 to 0.20), `n_estimators` (100 to 350),

`subsample` (0.6 to 1.0), `colsample_bytree` (0.6 to 1.0), `min_child_weight` (1 to 8), and L1/L2 regularization (`reg_alpha`, `reg_lambda`).

6.3 Comprehensive Model Leaderboard

Model Architecture	Imbalance / Tuning Policy	5-Fold CV ROC-AUC	Test ROC-AUC	Test PR-AUC	Test F1	Test Recall	Test Precision	Test Accuracy	Brier Loss
Logistic Regression (Vanilla)	Default Unweighted	0.8472	0.8431	0.6407	0.5902	52.94%	66.67%	80.48%	0.1375
Logistic Regression (Class-Weighted)	Balanced Cost-Weighting	0.8471	0.8431	0.6418	0.6126	77.81%	50.52%	73.88%	0.1661
Logistic Regression (SMOTE)	Synthetic Resampling	0.8351	0.8401	0.6393	0.6203	78.61%	51.22%	74.45%	0.1668
Random Forest (Balanced)	Balanced Subsample Depth 8	0.8467	0.8466	0.6628	0.6353	77.54%	53.80%	76.37%	0.1560
XGBoost Classifier (Tuned)	**Optuna Tuned + Scale Pos Weight**	**0.8489**	**0.8470**	**0.6624**	**0.6312**	**81.02%**	**51.71%**	**74.88%**	**0.1627**
LightGBM Classifier (Tuned)	Optuna Tuned + Scale Pos Weight	0.8481	0.8456	0.6627	0.6275	81.28%	51.09%	74.38%	0.1643

6.4 MLflow Tracking System

All experimental parameters, metrics across cross-validation folds, test metrics, and model binaries were logged to an SQLite-backed MLflow registry (`sqlite:///mlflow.db`) under experiment `telco_churn_experimentation`.

6.5 Champion Selection Rationale

The tuned **XGBoost Classifier** was selected as the champion model based on four technical criteria:
1. **Highest Area Under ROC:** Achieved **0.8470** holdout ROC-AUC and **0.8489** Stratified 5-Fold CV ROC-AUC.
2. **Top-Tier Detection Recall:** Correctly identified **81.02%** of all true churners in the holdout test set (303 out of 374).
3. **Probability Calibration:** Achieved a low Brier Loss (**0.1627**), ensuring risk scores correspond directly to empirical churn probabilities.
4. **Native SHAP Compatibility:** Fast and exact TreeExplainer execution for real-time customer explanations.

- **Serialized Artifact:** `models/best_model.joblib`.
- **Formal Log Report:** `reports/experiment_log.md`.
- **Notebook Reference:** `notebooks/03_model_experimentation.ipynb`.

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7. Week 3 Task 2: Global & Local Explainability (SHAP Analysis)

SHAP (Shapley Additive Explanations) decomposes each customer prediction into additive feature attributions derived from cooperative game theory.

7.1 Global Feature Importance Ranking (`src/explain.py`)

Rank	Feature Attribute	Mean Absolute SHAP Value	Directional Impact on Churn
1	`Contract_Two year`	**0.5116**	Strong Negative (Significantly reduces churn risk)
2	`tenure`	**0.3586**	Strong Negative (Higher accumulated months reduces churn risk)
3	`Contract_One year`	**0.3238**	Negative (Substantial churn protection)
4	`MonthlyCharges`	**0.2389**	Positive (Higher bills trigger customer flight)
5	`InternetService_Fiber optic`	**0.1919**	Positive (Elevated price sensitivity on fiber tiers)
6	`TotalCharges`	**0.1162**	Negative (Cumulative payments reflect brand commitment)
7	`PaymentMethod_Electronic check`	**0.1132**	Positive (Manual payment friction elevates churn)
8	`services_count`	**0.1101**	Negative (Multi-service bundling creates account stickiness)
9	`clv`	**0.1019**	Non-Linear (Interaction of tenure and billing rate)
10	`charge_ratio`	**0.0912**	Positive (Recent bill increases provoke attrition)

7.2 Explainability Visualizations

- **Global Feature Importance Bar:** `reports/figures/shap_summary_bar.png`
- **Beeswarm Distribution Plot:** `reports/figures/shap_beeswarm.png`
- Shows that high tenure (red dots) shifts log-odds heavily to the left (negative, reducing churn).
- High monthly charges and Fiber Optic subscriptions (red dots) shift log-odds to the right (positive, elevating churn).
- **Local Waterfall Plots:**

- *High-Risk Churner:* `reports/figures/shap_waterfall_churner.png` (Customer with 2 months tenure, month-to-month contract, fiber optic internet, paying $90+/mo without tech support).
- *Loyal Retained Subscriber:* `reports/figures/shap_waterfall_retained.png` (Customer with 65 months tenure, two-year contract, auto-pay, and 5 bundled services).
- **Formal SHAP Report:** `reports/shap_report.md`.

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8. Week 3 Task 2: Segment Error Audit (High-Value Long-Tenure Churners)

8.1 The Core Business Question

> **Executive Inquiry:** *Does the model miss long-tenure, high-value churners — the costliest false negatives?*

In telecom retention, all False Negatives do not have equal financial impact:
- Missing a **1-month subscriber paying $25/month** *causes an annualized revenue loss of* **300**.
- Missing a **48-month subscriber paying $105/month** *destroys over* **5,000 in CLV**.

8.2 Empirical Audit: Long-Tenure High-Value Cohort (`tenure > 24m`, `MonthlyCharges >= $70`)

Metric	Empirical Evaluation Result	Business Interpretation
Total Churners in Segment	**85 accounts**	Premium loyal customer base in jeopardy
Correctly Flagged Churners (TP)	**56 accounts**	Routed to proactive high-touch retention
Missed Churners (FN)	**29 accounts**	Residual unintercepted attrition
Segment Detection Recall	**65.88%**	Captures nearly 2 out of 3 high-value churners
Total CLV Protected (TP)	**$203,724.70**	High-value enterprise equity saved
Total CLV Lost (FN)	**$153,045.80**	Residual dollar exposure

8.3 Tenure Cohort Confusion Breakdown

Auditing the 1,409 holdout test customers across tenure cohorts:
- **0–12 Months:** The model captured the vast majority of churners. False Negatives in this bucket (21 accounts) were driven by nascent tenure with insufficient historical transaction signals.
- **13–36 Months:** Recall exceeded **80%**, as contract expiration signals provided strong predictive markers.
- **37–72 Months:** Base churn rate dropped to $< 8\%$. The model successfully identified long-tenure churners through anomalous billing spikes (`charge_ratio > 1.25`).
- **Figure References:** `reports/figures/segment_confusion_by_tenure.png`, `reports/figures/clv_at_risk_by_error_type.png`.

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9. Cost-Sensitive Threshold Optimization & Retention Economics

9.1 The Flaw of Default 0.50 Probability Cutoff

Standard machine learning models classify positive cases if $P(churn) \geq 0.50$. In customer retention, this assumption is economically suboptimal because:

$$Cost(False\ Negative) \approx 800 \text{ to } 1{,}500 \quad \gg \quad Cost(False\ Positive) = \$35.00$$

Because missing a churner costs **25x to 40x more** than offering an unneeded discount, the operational threshold should be lowered to intercept more borderline churners.

9.2 Optimization Across All Probability Cutoffs ($T \in [0.05, 0.95]$)

- **Retention Offer Cost:** $35.00 per customer.
- **Intervention Save Rate:** 45% of contacted churners accept the renewal offer and stay for ≥ 12 months.
- **Gross Revenue Preserved:** $12 \times MonthlyCharges_i$ per retained subscriber.



9.3 Retention Policy Comparison Matrix

Operating Retention Policy	Threshold T	Customers Targeted	Churn Recall	Campaign Cost	Net Retention Savings	Business Profit Uplift
No Model (Status Quo)	N/A	0 (0%)	0.0%	*0.00*	0.00	Baseline
Blanket Discount Campaign	0.00	1,409 (100%)	100.0%	*49,315.00*	102,085.46	Inefficient (Margins Burned)
Default ML Threshold	0.50	586 (41.6%)	81.02%	*20,510.00*	100,951.93	Strong Predictive Base
Optimal Cost-Tuned Threshold	**0.30**	**804 (57.1%)**	**92.51%**	***28,140.00***	**110,156.16**	**+$9,204.23 Net Profit Over Default**

- **Key Takeaway:** Lowering the threshold to $T^{\wedge} = 0.30^{*}$ expands the targeted pool to 804 customers, increasing churn recall to **92.51%** and capturing an additional **+$9,204.23** in net profit after subtracting all campaign costs.
- **Optimization Curve:** `reports/figures/cost_sensitive_threshold_curve.png`.
- **Formal Writeup:** `reports/error_analysis_writeup.md`.

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10. Model Governance & Model Card Summary

Following the Mitchell et al. (2019) framework, the model card establishes governance protocols:

- **Intended Use:** Proactive 90-day churn intervention and outbound retention campaign routing.
- **Out-of-Scope Uses:** Credit scoring, debt collection enforcement, or discriminatory price discrimination.
- **Fairness & Demographic Audit:**
- Evaluated churn parity across gender (`Male` vs `Female` F1: 0.630 vs 0.632).
- Evaluated `SeniorCitizen` equity. The higher churn observed in senior citizens (41.7%) is explained by lack of technical support and contract types rather than demographic bias.
- **Production Maintenance:** Retrain quarterly to prevent concept drift from competitor pricing moves.
- **Full Model Card Document:** `reports/model_card.md`.

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11. Repository Artifacts, Code Architecture & Reproduction Guide

11.1 Project Modules & Deliverable Assets

Asset / File	Description	Output Artifacts
`src/features.py`	Data cleaning, feature engineering, and `ColumnTransformer`	Transformed feature arrays, `models/preprocessor.joblib`
`src/train.py`	Stratified 5-Fold CV, Optuna tuning, MLflow logging	`models/best_model.joblib`, `reports/experiment_log.md`
`src/explain.py`	SHAP TreeExplainer, error audit, and threshold tuning	`reports/shap_report.md`, `reports/error_analysis_writeup.md`
`src/generate_notebooks.py`	Automated notebook builder & headless execution engine	Pre-executed `notebooks/02_*.ipynb`, `03_*.ipynb`, `04_*.ipynb`

11.2 Reproduction Instructions

1. Setup Virtual Environment & Dependencies

```
# Navigate to repository
cd c:\Users\kjuge\Documents\jugendrakashyap\AI-ML-Project

# Activate virtual environment
.\.venv\Scripts\Activate.ps1

# Install requirements
pip install -r requirements.txt
```

2. Execute Training Pipeline & MLflow Logging

```
python src/train.py
```

Trains baseline models, runs Optuna Bayesian search, evaluates Stratified 5-Fold CV, logs runs into `sqlite:///mlflow.db`, and serializes `models/best_model.joblib`.

3. Execute Explainability & Error Audit Pipeline

```
python src/explain.py
```

Calculates SHAP values, generates high-resolution figures in `reports/figures/`, performs segment error auditing, optimizes the cost threshold, and exports all markdown audit reports.

4. Launch the MLflow Experiment Tracking Dashboard

```
mlflow ui --backend-store-uri sqlite:///mlflow.db
```

Access `http://localhost:5000` in your web browser to review model parameters, loss curves, metrics, and logged model signatures.

5. Interactive Jupyter Notebooks

```
jupyter notebook
```

The repository includes four self-contained notebooks:
- `notebooks/01_eda.ipynb`: Week 1 EDA & Data Quality Report
- `notebooks/02_features_and_baseline.ipynb`: Week 2 Feature Engineering & Logistic Regression Baseline
- `notebooks/03_model_experimentation.ipynb`: Week 3 Model Experimentation & MLflow Logging
- `notebooks/04_explainability_and_error_analysis.ipynb`: Week 3 SHAP Interpretability & Cost-Sensitive Tuning